

Machine Learning (EE526/RA506)

Mid-Semester Exam

Date: 25/02/2024

Time: Two Hours

Full Marks: 100

[Q1] Consider the following $2d$ points $X_i = [x_i, y_i]^T$; $i = 1, \dots, 5$.

x	-1.2	5.2	-4.2	5.1	-9.4
y	7.3	-6.5	1.3	-8.2	3.5

[a] Compute the centroid μ and Covariance Matrix C .

[2+5=7]

[b] Consider the $2d$ Gaussian distribution $G(X) = \eta \exp\{(X - \mu)^T A^{-1}(X - \mu)\}$ whose contour plot is a set of oriented concentric ellipses, centered at μ with major and minor axes along the eigen vectors of C . Evaluate η and A .

[2+1=3]

[c] Evaluate the ratio of spreads along major-to-minor axis directions of these oriented ellipses.

[5]

[Q2] Consider the problem of model fitting using regression diagnostics on a noisy data set containing α percentage of outliers. We use λ as the Chebyshev inequality threshold for inlier-outlier classification in every iteration of regression diagnostics algorithm. Deduce the approximate number of regression diagnostics iterations required for rejecting all the outliers. Suppose the problem needs m number of points to fit the model. Deduce the expected number of RANSAC trials required for such a problem.

[12+5=17]

[Q3] We are minimizing the objective function $g(x) = \frac{(x-110)^2}{100}$ using stochastic search with a solution population of size 6. Consider the following parent solution population at the iteration t as $S_t = \{76, 82, 132, 155, 100, 124\}$. Find the number of children that will be generated from each of these solutions using the process of random walk. Report the result in form of the following table. [3 × 6 = 18]

Parent Solution	76	82	132	155	100	124
No. of Children						

[Q4] The dataset $S = \{x_i; x \in \mathbb{R}^d; i = 1, \dots, n\}$ ($n = 210000, d = 10$) is indexed by using K-D Tree.

This is realized through a fully-grown binary tree which terminates with at least $m = 100$ points at each leaf

[Q5] Let n_i , μ_i and σ_i be the respective size (number of points), mean and standard deviation of the dataset Z_i ($i = 1, 2$). The dataset Z_i is obtained by projecting d -dimensional points of dataset D_i along the unit direction vector ω .

The direction vector ω is to be optimally chosen for maximum discrimination of the projected pointsets Z_1 and Z_2 . The objective function for this task is as follows.

Deduce the expression for ω that maximizes $J(\omega)$. [15]

2	36	61	5	33	4	14	29	1	63
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[illegible]